

Yutian Chen

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Education

Stanford University, PhD. Computer Science	Sep 2026 - Now
Carnegie Mellon University, MS. Robotics (Dropped out)	Sep 2025 - May 2026
Carnegie Mellon University, BSc. Computer Science	Sep 2021 - May 2025

Experience

Member of Technical Staff Intern , Amazon Frontier AI Robotics	May 2026 - Aug 2026
Working on IMU foundation model for humanoid and human motion as a robust and generalizable encoder for multiple downstream tasks.	
Research Assistant , Prof. Sebastian Scherer, Carnegie Mellon University	Sep 2022 - May 2026
- Co-proposed the <i>MAC-VO</i> idea and developed the full system from scratch, including the core stereo VO design and implementation, data and training infrastructure, and experimental evaluation. - Built and deployed real-time perception stacks on NVIDIA Jetson Orin and Thor, designing a multi-process, GPU-accelerated pipeline with ROS and TensorRT, making <i>MAC-VO</i> and <i>UFM</i> meets compute limit for UAV. - Developed inertial odometry and uncertainty modules (<i>AirIO</i>, <i>AirIMU</i>), implementing differentiable Euler/RK4 IMU pre-integration, train and evaluation pipelines across platforms, and integrating them into the SLAM stack.	
Research Scientist Intern , Field AI	Jun 2025 - Aug 2025
- Led the research of <i>Co-Me</i>, a confidence-guided token-merging method that accelerates VGGT and related visual geometric transformers by up to $11.3\times$ without retraining, enabling real-time 3D perception for embodied AI. - Built the end-to-end training, evaluation, and profiling pipeline for <i>Co-Me</i>, wrote custom CUDA kernels and verified the acceleration on both datacenter GPUs and Jetson-class edge compute.	
Research Assistant , Prof. Chuang Gan, MIT-IBM Watson AI Lab	Apr 2024 - Jan 2025
- Key contributor to <i>Virtual Community</i> project, developed a city-scale 3D scene generation pipeline from satellite, aerial, and street-view imagery, including reconstruction, meshing, and refinement for urban environment. - Integrated this pipeline with the <i>Genesis</i> simulation engine to support large-scale physics simulation, enabling scalable evaluation of visuomotor agents in real-world grounded scenes.	
Research Assistant , Prof. Bhiksha Raj, Carnegie Mellon University	Mar 2023-Sep 2023
- Designed and implemented an LLM-generated content detector based on token-prediction signals, including model architectures, training pipeline, and large-scale evaluation across diverse domains. - Curated and released the <i>OpenLLMText</i> dataset (human texts with LLM rewrites), defining data collection, filtering, and quality-control procedures and packaging tools for public use.	

Publications

MAC-VO: Metrics-aware Covariance for Learning-based Stereo Visual Odometry	ICRA 2025
Yuheng Qiu*, Yutian Chen *, Zihao Zhang, Wenshan Wang, Sebastian Scherer	Best Conference & Perception Paper
https://arxiv.org/abs/2409.09479	
Co-Me: Confidence-Guided Token Merging for Visual Geometric Transformers	CVPR 2026
Yutian Chen , Yuheng Qiu, Ruogu Li, Jay Patrikar, Sebastian Scherer	
https://arxiv.org/abs/2511.14751	
UFM: A Simple Path towards Unified Dense Correspondence with Flow	NeurIPS 2025
Yuchen Zhang, Nikhil Keetha, Chenwei Lyu, Bhuvan Jhamb, Yutian Chen , Yuheng Qiu, Jay Karhade, Shreyas Jha, Yaoyu Hu, Deva Ramanan, Sebastian Scherer, Wenshan Wang	

<https://arxiv.org/abs/2506.09278>

AirIO: Learning Inertial Odometry with Enhanced IMU Feature Observability

IEEE RA-L

Yuheng Qiu*, Can Xu*, **Yutian Chen**, Shibo Zhao, Junyi Geng, Sebastian Scherer
<https://arxiv.org/abs/2501.15659>

AirIMU: Learning Uncertainty Propagation for Inertial Odometry

arXiv Preprint

Yuheng Qiu, Chen Wang, Can Xu, **Yutian Chen**, Xunfei Zhou, Youjie Xia,
Sebastian Scherer
<https://arxiv.org/abs/2310.04874>

Virtual Community: An Open World for Humans, Robots, and Society

ICLR 2026

Qinhong Zhou*, Hongxin Zhang*, Xiangye Lin*, Zheyuan Zhang*, **Yutian Chen**,
Wenjun Liu, Zunzhe Zhang, Sunli Chen, Lixing Fang, Qiushi Lyu, Xinyu Sun,
Jincheng Yang, Zeyuan Wang, Bao Chi Dang, Zhehuan Chen, Daksha Ladia,
Jiageng Liu, Chuang Gan
<https://arxiv.org/abs/2508.14893>

Token Prediction as Implicit Classification to Identify LLM-Generated Text

EMNLP 2023

Yutian Chen*, Hao Kang*, Vivian Zhai, Liangze Li, Rita Singh, Bhiksha Raj
<https://aclanthology.org/2023.emnlp-main.810/>

PyPose v0.6: The Imperative Programming Interface for Robotics

IROS Workshop, 2023

Zitong Zhan, Xiangfu Li, . . . , **Yutian Chen**, . . . , Jiajun Wu, . . . , Chen Wang
<https://arxiv.org/abs/2309.13035>

Myocardial Segmentation of Cardiac MRI Sequences With Temporal Consistency for Coronary Artery Disease Diagnosis

Frontier Cardiovascular
Medicine, 2022

Yutian Chen, Wen Xie, Jiawei Zhang, Hailong Qiu, Dewen Zeng, Yiyu Shi,
Haiyun Yuan, Jian Zhuang, Yanchun Zhang, Yuhao Dong, Meiping Huang,
Xiaowei Xu
<https://www.frontiersin.org/articles/10.3389/fcvm.2022.804442/full>

Teaching

Teaching Assistant, Intro to Machine Learning (PhD), Carnegie Mellon

Sep 2023 - Jan 2024

- Led 80-minute recitation lectures, prepared course material and assignments, held office hours.

Teaching Assistant, Principle of Imperative Programming, Carnegie Mellon

Sep 2022 - Jan 2023

- Led lab session, developed a virtual machine for C-like program heap visualization on browser environments.

Projects

Perceptual Local-manipulation for Humanoid

Oct 2025 - Jan 2026

- Extending the *HDML: Learning Interactive Humanoid Whole-Body Control from Human Videos* by removing privileged state inputs and replacing them with ego-centric perception with uncertainty from *MAC-VO*.
- Designing a visual state-estimation front-end that infers humanoid pose, contact state, and scene geometry directly from stereo images, feeding these estimates into the whole-body control policy.

NASA Lunar Autonomy Challenge (Team Lead, 3rd Place)

Sep 2024 - May 2025

- Managed the full development lifecycle for the NASA Lunar Autonomy Challenge, coordinating efforts across perception, planning, and control teams. Orchestrated the system integration and testing of autonomy stack.
- Led the development of visual-inertial SLAM system with less than 0.1cm/s drift in extraterrestrial environment.

Latch-free Parallel B+ Tree

Mar 2024 - Apr 2024

- Designed and implemented an efficient latch-free B+ tree data structure using C++ 17. Utilized SIMD, multiprocessing and CAS operation on x86-64 architecture to achieve 20M read/write QPS on laptop CPU.